

Day 9 (11/11/2025)

Q) Write a Program to check the numbers from 1 to 10 is even or odd.

1 - odd
2 - even
3 - odd
4 - even . . .

o/p

1 - odd
2 - even
3 - odd
4 - even
5 - odd
6 - even
7 - odd
8 - even
9 - odd
10 - even

```
A) for i in range(1,11,1):  
    if i%2==0:  
        print(i, "- even")  
    else:  
        print(i, "- odd")
```

Q) Write a Program to calculate 1-10
the sum of numbers 55
the ~~even~~ sum of even numbers 30
sum of odd numbers 25
from 1 to 10

```
ts = 0  
es = 0  
os = 0  
for i in range(1,11,1):
```

```
    ts = ts + i  
    if (i%2==0):  
        es = es + i  
    else:  
        os = os + i
```

```
print(f"total sum = {ts} \neven sum = {es} \n odd sum = {os}")
```

o/p

total sum = 55
even sum = 30
odd sum = 25

Q) Write a Program to print tables of given number;

o/p

$$3 \times 1 = 3$$

$$3 \times 2 = 6$$

$$3 \times 3 = 9$$

...

$$3 \times 10 = 30$$

i/p $\Rightarrow 3$

$$\text{o/p} \Rightarrow 3 \times 1 = 3$$

$$3 \times 2 = 6$$

$$3 \times 3 = 9$$

$$3 \times 4 = 12$$

$$3 \times 5 = 15$$

$$3 \times 6 = 18$$

$$3 \times 7 = 21$$

$$3 \times 8 = 24$$

$$3 \times 9 = 27$$

$$3 \times 10 = 30$$

a) $n = \text{int}(\text{input}())$

for i in range(1, 11, 1):

print(f"{n} X {i} = {n*i}")

Q) Write a Program to calculate factorial of given number.

$n = \text{int}(\text{input}())$ # increment method

$$\Rightarrow 5 \times 4 \times 3 \times 2 \times 1 \Rightarrow 120$$

f = 1

i/p $\Rightarrow 5$

o/p $\Rightarrow$ factorial of given number = 120

for i in range(1, n+1, 1):

f = f * i

print("factorial of given number:", f)

$n = \text{int}(\text{input}())$ # decrement method

i/p $\Rightarrow 5$

o/p $\Rightarrow$ factorial of given number = 120

f = 1

for i in range(n, 0, -1):

f = f * i

print("factorial of given num:", f)

While loop - While loop is used to execute a block of statements repeatedly until a given condition is satisfied. When the condition becomes false, the line immediately after the loop in the program is executed.

while loop Syntax.

initialization while

while (condition):

Statements of while

incr/decr of while.

a) Print first ten numbers

i = 1

while (i <= 10):

print(i)

i += 1

o/p)

1
2
3
4
5
6
7
8
9
10

print Python in vertical

name = "python"

i = 0

while (i < len(name)):

print(name[i])

i = i + 1

o/p)

p
y
t
h
o
n

a) print position & value of python.

i = 0

p = 0

while (i < len(name)):

print(f"pos = {p} , value = {name[i]}")

i = i + 1

p = p + 1

o/p) pos = 0, value = p

pos = 1, value = y

pos = 2, value = t

pos = 3, value = h

pos = 4, value = o

pos = 5, value = n

Q) Print "Hello" 10 times.

i = 1

while (i <= 10):

print("i =", i)

print("Hello")

i = i + 1

o/p> Hello

i = 2

Hello

i = 3

Hello

i = 4

Hello

i = 5

Hello

i = 6

Hello

i = 7

Hello

i = 8

Hello

i = 9

Hello

i = 10

Hello

Q) Print 1 to 20 numbers.

i = 1

while (i <= 20):

print(i)

i = i + 1

o/p>

1
2
3
4
5
6
7
8
9
10

11
12
13
14
15
16
17
18
19
20

Q) print even & odd numbers from 1 to 10.

i = 0

while (i <= 10):

if (i % 2 == 0):

print("even =", i)

else:

print("odd =", i)

i = i + 1

even = 0

odd = 1

even = 2

odd = 3

even = 4

odd = 5

even = 6

odd = 7

even = 8

odd = 9

even = 10

Q) " write a Program to calculate the sum of numbers 55, sum of even numbers 30, sum of odd 25 from 1 to 10 "

ts = 0

es = 0

os = 0

i = 0

while (i <= 10):

ts = ts + i

if (i % 2 == 0):

es = es + i

else:

os = os + i

i = i + 1

o/p>

total sum = 55

even sum = 30

odd sum = 25

print(f"total sum = {ts} | even sum = {es} | odd sum = {os}")

Q) write a program to print tables of given number

''' o/p
3 x 1 = 3
3 x 2 = 6
1'...'''
3 x 10 = 30''

```
n = int(input())
```

```
i = 1
```

```
while (i <= 10):
```

```
    print(f"{n} x {i} = {n*i}")
```

```
    i = i + 1
```

i/p = 3

o/p $\Rightarrow$

3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

Q) Write a Program to calculate factorial of given number.

```
n = int(input("enter a number = "))
```

i/p = 5

o/p factorial of given num = 120

```
i = 1
```

```
f = 1
```

```
while (i <= n):
```

```
    f = f * i
```

```
    i = i + 1
```

```
print("factorial of given number = ", f)
```